



BMC
TRAINING

PROFESSIONAL TRAINING PROGRAMME

Access Control and Biometric Systems Administration

Three-day instructor-led programme

A practical programme on administering, monitoring, auditing and maintaining electronic access control and biometric systems, from readers, controllers and credentials through to user enrolment, access-log review, troubleshooting and the responsible protection of biometric information.

DURATION 3 days	LEVEL Intermediate practical	FORMAT Instructor-led
APPROACH Hands-on demos & case studies	STRUCTURE 17 modules	CERTIFICATE Certificate of Attendance

OVERVIEW**Course introduction**

Electronic access control and biometric security systems have become essential components of modern physical security. Banks, mines, industrial facilities, data centres, government buildings, airports, hospitals, corporate offices and other high-security environments increasingly depend on electronic access systems to control who may enter restricted areas, when access is permitted and how access activities are recorded.

However, installing access control technology alone does not guarantee effective security. Poor system administration can result in inactive employees retaining access rights, duplicated or “ghost” users remaining on databases, inappropriate access privileges, weak enrolment practices, unmonitored access events and inadequate protection of sensitive biometric information.

This course provides delegates with practical knowledge and skills for administering, monitoring, auditing and maintaining electronic access control and biometric systems. It covers the complete administrative lifecycle, from understanding readers, controllers and credentials through to designing access zones, configuring schedules and permissions, enrolling users, managing biometric templates, reviewing access logs, investigating abnormal activity and maintaining system availability.

Delegates also explore the integration of access control with alarms, turnstiles, visitor management and other physical security systems. Particular attention is given to the responsible collection, storage, retention and protection of biometric information in accordance with applicable privacy and data-protection principles.

Practical exercises and realistic scenarios are incorporated throughout the three days so that delegates can apply the concepts to security environments within their own organisations.

OBJECTIVES**Course objectives**

By the end of this three-day course, participants should be able to:

- Explain the architecture and operating principles of modern electronic access control systems.
- Identify the functions of controllers, readers, credentials, biometric devices, locks, databases and management software.
- Configure logical and physical access zones according to organisational security requirements.
- Establish time schedules, access groups and permission levels.
- Apply the principle of least privilege when assigning physical access rights.
- Enrol and administer users using cards, PINs and biometric credentials.
- Modify, suspend and revoke user access efficiently.
- Establish appropriate processes for employee transfers, contractors and terminated personnel.
- Identify and remove duplicate, expired, inactive and ghost users.
- Manage biometric enrolment and templates securely.
- Review access logs and system events for suspicious or abnormal behaviour.
- Identify indicators of credential sharing, repeated denied access, attempted unauthorised entry and other access anomalies.
- Understand measures used to reduce tailgating and unauthorised entry.
- Integrate access control with alarms, turnstiles, visitor management and related security systems.
- Apply appropriate privacy and data-protection principles to biometric information.
- Establish appropriate controls for biometric data retention, access and deletion.
- Conduct routine maintenance of readers, controllers and system databases.
- Apply structured troubleshooting techniques to common access control problems.
- Develop practical access-control administration and audit procedures for their organisation.

AUDIENCE**Who should attend**

This programme is suitable for personnel responsible for the administration, operation, maintenance, supervision or oversight of electronic physical-security systems, including:

- Access Control Administrators
- Security Systems Administrators
- Security Systems Operators
- Control Room Operators and Supervisors
- Physical Security Officers
- Security Managers and Supervisors
- Facilities Managers
- Facilities and Maintenance Technicians
- Electronic Security Technicians
- ICT and Systems Support Personnel supporting physical security systems
- Biometric System Administrators
- Visitor Management Personnel
- Loss Prevention and Risk Personnel
- Compliance and Security Audit Personnel
- Personnel responsible for employee and contractor access permissions

The programme is also suitable for organisations seeking to strengthen the administration and governance of existing access control and biometric security infrastructure.

PROGRAMME

Detailed three-day agenda**DAY 1** Access Control Architecture, Configuration and User Administration**Module 1. Fundamentals of Electronic Access Control**

- Purpose and principles of physical access control
- Moving from traditional keys to electronic access systems
- Authentication, authorisation and accountability
- Identification versus verification
- Understanding “who, where and when” access rules
- Overview of card, PIN, mobile and biometric access systems
- Single-factor and multi-factor authentication
- Physical versus logical access control
- Common access-control applications in high-security facilities

Module 2. Access Control System Components

- Access control management software
- System servers and databases
- Intelligent controllers and control panels
- Door controllers and input/output modules
- Card and credential readers
- Biometric readers and terminals
- Electric strikes, magnetic locks and other locking devices
- Request-to-exit devices
- Door-position sensors
- Emergency release mechanisms
- Power supplies and backup power
- Network connectivity and communication
- Understanding the interaction between system components
- Basic system architecture and communication flow

Module 3. Access Zones and Security Area Design

- Identifying security zones within a facility
- Public, controlled, restricted and high-security areas
- Mapping doors and access points to security zones
- Designing access according to operational requirements
- Separation of duties and sensitive areas
- High-risk locations and critical infrastructure
- Anti-passback concepts
- Mantrap and interlocking door concepts
- Emergency and evacuation considerations
- Developing a practical access-control matrix

Module 4. Time Schedules and Permission Levels

- Creating time zones and access schedules
- Normal working hours
- Shift-based access
- Weekend and public-holiday access
- Temporary access periods
- Restricted after-hours access
- Access groups and permission profiles
- Role-based access assignment
- Applying least-privilege principles
- Avoiding excessive access rights
- Managing exceptions and special permissions
- Periodic access-right reviews

Module 5. User Enrolment and Credential Administration

- User identity verification before enrolment
- Creating user records
- Card enrolment and assignment
- PIN administration
- Biometric enrolment
- Assigning departments, roles and access groups
- Employee versus contractor credentials
- Temporary credentials
- Lost, damaged and stolen credentials
- Suspending and reactivating users
- Employee transfer and role-change procedures
- Termination and immediate revocation
- Preventing duplicate and ghost user accounts
- Maintaining accurate user databases

Day 1 practical exercise. Participants work through a simulated facility and:

- Identify security zones
- Develop an access matrix
- Create user categories
- Assign appropriate permission levels
- Establish access schedules
- Determine appropriate access for employees, contractors and visitors
- Identify potential weaknesses in an incorrectly configured access structure

DAY 2 Biometrics, Monitoring, Auditing and System Integration**Module 6. Biometric Access Control Technologies**

- Introduction to biometric authentication
- Fingerprint recognition
- Facial recognition
- Iris and other biometric technologies
- Biometric templates versus raw biometric images
- Enrolment processes
- Quality of biometric enrolment
- Factors affecting biometric performance
- False acceptance and false rejection
- Matching thresholds
- Environmental considerations
- Dealing with users who cannot reliably enrol
- Combining cards, PINs and biometrics
- Multi-factor authentication for high-security areas

Module 7. Biometric User Management

- Establishing biometric enrolment procedures
- Identity validation before enrolment
- Capturing high-quality biometric data
- Linking biometric templates to user profiles
- Updating biometric records
- Removing biometric access
- Managing employees, contractors and visitors
- Handling failed biometric authentication
- Preventing unauthorised enrolment
- Administrative privileges for biometric systems
- Maintaining enrolment and deletion records

Module 8. Access Record Monitoring and Audit Trails

- Understanding access-control event logs
- Successful access events
- Access-denied events
- Door-held-open events
- Door-forced-open events
- Invalid credentials
- Repeated authentication failures
- After-hours activity
- Access attempts at unusual locations
- Reviewing administrator activity
- Importance of accurate system time
- Retaining access records
- Producing security and audit reports

Module 9. Detecting Access Control Anomalies

- Identifying ghost users
- Detecting expired or inactive accounts
- Credential-sharing indicators
- Abnormal access patterns
- Repeated denied access
- Attempts to access unauthorised zones
- Suspicious after-hours activity
- Simultaneous or geographically impossible access events
- Tailgating and piggybacking risks
- Anti-passback as a detection and control measure
- Escalating suspicious events
- Preserving records for investigation
- Establishing regular access-control audits

Module 10. Integration with Other Security Systems

- Access control and intrusion alarms
- Access control and CCTV/video surveillance
- Turnstile integration
- Boom gates and vehicle access
- Visitor management systems
- Intercom and entry systems
- Fire alarm and emergency-release considerations
- Linking access events with security alarms
- Automatic event notification
- Central security control rooms
- Benefits and risks of integrated security platforms

Module 11. Visitor and Contractor Access Management

- Pre-registration procedures
- Identity verification
- Temporary credentials
- Escort requirements
- Restricted visitor zones
- Contractor access periods
- Managing recurring contractors
- Visitor access expiration
- Credential return and deactivation
- Maintaining visitor records
- Managing access outside normal working hours

Day 2 practical exercise. Participants analyse a simulated access-control event log containing:

- Repeated denied-access attempts
- After-hours access
- Expired credentials
- Possible credential sharing
- Ghost user accounts
- Door-forced-open events
- Unusual access patterns

Delegates identify the security concerns, prioritise the events and recommend appropriate administrative and investigative actions.

DAY 3 Biometric Data Protection, Maintenance, Troubleshooting and Security Governance**Module 12. Biometric Data Privacy and Protection**

- Why biometric information requires enhanced protection
- Understanding personal and sensitive information
- Privacy principles applicable to biometric systems
- Purpose limitation
- Data minimisation
- Transparency and appropriate notification
- Lawful and responsible processing
- Access controls for biometric databases
- Secure storage of biometric templates
- Transmission and network security considerations
- Retention periods
- Secure deletion and disposal
- Third-party biometric service providers
- Responding to suspected biometric data breaches
- Maintaining records demonstrating responsible administration
- Aligning organisational procedures with applicable privacy and data-protection legislation

Module 13. System Administration Security

- Protecting administrative accounts
- Role-based administrator permissions
- Limiting privileged access
- Password and authentication controls
- Managing vendor and technician access
- Administrator activity logs
- Configuration change records
- Software and firmware management
- Backups and database protection
- Network-security considerations
- Preventing unauthorised configuration changes
- Periodic administrator account reviews

Module 14. Access Control System Maintenance

- Importance of preventive maintenance
- Routine inspection of readers
- Checking door-position sensors
- Testing locks and release mechanisms
- Controller health checks
- Power-supply inspection
- Backup battery testing
- Network and communication checks
- Reader cleaning and environmental maintenance
- Database maintenance
- System backups
- Reviewing storage capacity and event logs
- Firmware and software updates
- Maintaining equipment and service records
- Establishing maintenance schedules

Module 15. Troubleshooting Common Access Control Problems

- Structured troubleshooting methodology
- User credential not recognised
- Card reader not responding
- Biometric reader rejecting valid users
- Door not unlocking
- Door remaining unlocked
- Controller communication failure
- Network connectivity problems
- Incorrect access schedules
- Permission and access-group errors
- Database synchronisation problems
- Power-related faults
- Intermittent device problems
- Identifying whether a problem is hardware, software, configuration or user related
- Recording faults and corrective actions
- Escalating faults to vendors or technical specialists

Module 16. Access Control Auditing and Governance

- Purpose of periodic access reviews
- User-account reconciliation
- Reviewing privileged users
- Reviewing contractors and temporary accounts
- Checking terminated employee access
- Comparing HR records with access-control databases
- Reviewing high-security permissions
- Audit-log review
- Administrator activity review
- Documenting exceptions
- Corrective action tracking
- Developing an access-control audit checklist
- Establishing management reporting procedures

Module 17. Incident Response and Access Control Investigations

- Responding to unauthorised access attempts
- Investigating suspicious access events
- Preserving system logs
- Correlating access events with CCTV and alarm information
- Identifying affected doors, users and time periods
- Credential suspension during investigations
- Escalation and notification
- Documenting investigation findings
- Corrective and preventive action
- Lessons learned after access-control incidents

Day 3 practical exercise: Access control security review. Participants conduct a structured review of a simulated organisation's access-control environment, including:

- Reviewing user accounts
- Identifying ghost and inactive users
- Reviewing access permissions
- Checking privileged administrator accounts
- Examining biometric administration practices
- Reviewing suspicious access events
- Identifying maintenance weaknesses
- Assessing privacy and data-protection risks
- Recommending corrective actions

Participants then prepare a short Access Control Improvement Action Plan setting out priority risks, corrective actions and recommended administrative controls.

APPROACH

Training methodology

The programme uses an interactive and practical training approach combining:

- Instructor-led presentations
- Practical demonstrations
- Realistic access-control scenarios
- Configuration exercises
- Access-log analysis
- Group discussions
- Security case studies
- Troubleshooting exercises
- Access-control audit exercises
- Question-and-answer sessions

Hands-on practice. Where suitable equipment or demonstration systems are available, practical activities may include reader configuration, user enrolment, access-level assignment, biometric enrolment, event-log review and basic troubleshooting.

CERTIFICATION

Certificate

Participants who complete the programme will receive a BMC Training Certificate of Attendance confirming participation in the Access Control and Biometric Systems Administration course.

OUTCOME

Key takeaways

Upon successful completion of the three-day programme, delegates should be better equipped to administer an electronic access-control environment from initial user enrolment through daily monitoring, maintenance, audit and eventual revocation of access. Participants should also be able to recognise weaknesses that can lead to unauthorised access, improve the quality of user and permission administration, investigate abnormal access events and apply stronger controls to the handling of biometric information.

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